

Second Midterm Exam

Name: _____

Please circle your section number!

SECTION 1

SECTION 2

SECTION 3

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(MWF 9:55 AM)

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(TR 9:30 AM)

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(MWF 8:50 AM)

- There are 6 problems on the exam, some of them have multiple parts.
- Read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.**
- You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.

Question	Points	Score
1	15	
2	5	
3	10	
4	10	
5	12	
6	8	
Total:	60	

1. Let X be a random variable with probability density function

$$f_X(x) = \begin{cases} x^{-2}, & x \geq 1; \\ 0, & \text{otherwise.} \end{cases}$$

(a) (8 points) (3+5) Define a discrete random variable W by setting

$$W = \left\lfloor \frac{3}{X} \right\rfloor.$$

(Here, $\lfloor \cdot \rfloor$ indicates the floor function, defined by $\lfloor t \rfloor = \max\{n \in \mathbb{Z} : n \leq t\}$.)

Let $a = P(W = 0)$, let $b = P(W = 1)$, and let $c = P(W = 2)$.

i Determine the numerical values of a , b , and c .

Solution:

We have that $W = 0 \iff X > 3$, so

$$a = P(X > 3) = \int_3^\infty f_X(x) dx = \int_3^\infty x^{-2} dx = (-x^{-1}) \Big|_{x=3}^\infty = 0 - \left(-\frac{1}{3}\right) = \frac{1}{3}.$$

Meanwhile, $W = 1 \iff \frac{3}{2} < X \leq 3$, so

$$b = \int_{3/2}^3 f_X(x) dx = \int_{3/2}^3 x^{-2} dx = (-x^{-1}) \Big|_{x=3/2}^3 = -\frac{1}{3} - \left(-\frac{2}{3}\right) = \frac{1}{3}.$$

Finally, $W = 2 \iff 1 < X \leq \frac{3}{2}$, so

$$c = \int_1^{3/2} f_X(x) dx = \int_1^{3/2} x^{-2} dx = (-x^{-1}) \Big|_{x=1}^{3/2} = -\frac{2}{3} - (-1) = \frac{1}{3}.$$

ii Determine the moment generating function of W , and use this to compute the expectation and variance of W . (To receive full credit you must compute expectation and variance by differentiating the moment generating function. Even if you do not solve part (i), you can get full credit for correct answers in terms of a , b , and c .)

Solution: In terms of a , b , and c , the moment generating function of W and its first two derivatives are as follows:

$$M_W(t) = a + be^t + ce^{2t};$$

$$M'_W(t) = be^t + 2ce^{2t};$$

$$M''_W(t) = be^t + 4ce^{2t}.$$

If we substitute $a = b = c = \frac{1}{3}$, then this becomes

$$M_W(t) = \frac{1}{3} (1 + e^t + e^{2t});$$

$$M'_W(t) = \frac{1}{3} (e^t + 2e^{2t});$$

$$M''_W(t) = \frac{1}{3} (e^t + 4e^{2t}).$$

In terms of b and c , the expectation and variance of W are given by

$$\begin{aligned} E[W] &= M'_W(0) = be^0 + 2ce^{2(0)} = b + 2c; \\ \text{Var}(W) &= E[W^2] - (E[W])^2 = M''_W(0) - (M'_W(0))^2 \\ &= (b + 4c) - (b + 2c)^2 = b + 4c - (b^2 + 4bc - 4c^2) \end{aligned}$$

Meanwhile, using the second set of equations, we obtain numerical answers as follows:

$$\begin{aligned} E[W] &= M'_W(0) = \frac{1}{3} (e^0 + 2e^{2(0)}) = 1; \\ \text{Var}(W) &= M''_W(0) - (M'_W(0))^2 = \frac{1}{3}(1 + 4) - \frac{1}{9}(1 + 2)^2 = \frac{5}{3} - 1 = \frac{2}{3}. \end{aligned}$$

(b) (7 points) Define a continuous random variable Y by setting

$$Y = \log(X),$$

(Here, as before, X has density function

$$f_X(x) = \begin{cases} x^{-2}, & x \geq 1; \\ 0, & \text{otherwise.} \end{cases}$$

Meanwhile, $\log$ indicates the natural logarithm.) Determine the cumulative distribution function (CDF) of Y , and use your answer to find the probability density function (PDF) of Y .

Solution: We first find the cumulative distribution function of X . For $x < 1$, we have that $F_X(x) = 0$. Meanwhile, for $x \geq 1$, we have that

$$F_X(x) = \int_1^x t^{-2} dt = (-t^{-1}) \Big|_{t=1}^x = 1 - \frac{1}{x}.$$

Overall,

$$F_X(x) = \begin{cases} 1 - \frac{1}{x}, & x \geq 1; \\ 0, & x < 1. \end{cases} \quad (1)$$

Now, we determine the cumulative distribution function of Y . For each $y \in \mathbb{R}$, we have that

$$F_Y(y) = P(Y \leq y) = P(\log X \leq y) = P(X \leq e^y) = F_X(e^y).$$

From (1), combined with the observation that $e^y \geq 1 \iff y \geq 0$, we see that

$$F_Y(y) = \begin{cases} 1 - e^{-y}, & y \geq 0; \\ 0, & y < 0. \end{cases} \quad (2)$$

Finally, we differentiate (2) to obtain

$$f_Y(y) = \begin{cases} e^{-y}, & y \geq 0; \\ 0, & y < 0. \end{cases}$$

Alternate Solution: The above was the intended solution. However, we also gave full credit if you correctly applied the formula

$$f_Y(y) = f_X(g^{-1}(y)) \left(\frac{1}{|g'(g^{-1}(y))|} \right) \quad (3)$$

for $g(x) = \log x$ and *also found the CDF of Y by integrating*. Specifically, applying (3) gives

$$f_Y(y) = \begin{cases} \frac{1}{(e^y)^2} \left(\frac{1}{1/e^y} \right), & e^y \geq 1; \\ 0, & e^y < 1 \end{cases} = \begin{cases} e^{-y}, & y \geq 0; \\ 0, & y < 0 \end{cases}.$$

Since $f_Y(y) = 0$ for $y < 0$, it follows that $F_Y(y) = 0$ for $y < 0$. Meanwhile, for $y \geq 0$, we have

$$F_Y(y) = \int_0^y f_Y(s) ds = \int_0^y e^{-t} dt = (-e^{-t}) \Big|_{t=0}^y = 1 - e^{-y}.$$

Overall,

$$F_Y(y) = \begin{cases} 1 - e^{-y}, & y \geq 0; \\ 0, & y < 0; \end{cases}$$

2. Andrew writes a very popular webcomic and posts new panels regularly. Some panels have black and white illustrations, and some panels have illustrations in color. Suppose that the number X of new panels in a given week is $\text{Poisson}(\lambda)$ for some λ .

- (a) (2 points) Suppose that the probability that the webcomic has no new panels in a given week is $\frac{4}{13}$. Determine the value of λ .

Solution: Using the probability mass function for a $\text{Poisson}(\lambda)$ random variable, we see that

$$\frac{4}{13} = P(X = 0) = e^{-\lambda} \left(\frac{\lambda^0}{0!} \right) = e^{-\lambda}.$$

Thus,

$$\lambda = \ln \left(\frac{13}{4} \right).$$

- (b) (3 points) Let Y be the number of panels in a given week that have color illustrations. Suppose that a panel has color illustrations with probability $\frac{1}{3}$, and that the presence of color in one panel is independent of the presence of color on another panel. Write (but do not evaluate) a sum that gives the probability $P(Y = 4)$.

Hint: it may be helpful to use the law of total probability and condition on the value of X .

Solution: Since the presence of color in one panel is independent of the presence of color in another panel, it follows that for a fixed $n \in \{0, 1, 2, 3, \dots\}$, the number of panels, out of n , that have color illustrations is binomially distributed. For $n \geq 4$, the conditional probability $P(Y = 4|X = n)$ is given by

$$P(Y = 4|X = n) = \binom{n}{4} \left(\frac{2}{3} \right)^{n-4} \left(\frac{1}{3} \right)^4.$$

(In fact, this formula is also valid for $n \in \{0, 1, 2, 3\}$ but would give 0.)

Using the law of total probability, we multiply the above by $P(X = n)$ and sum over $n \geq 4$ to obtain

$$\begin{aligned} P(Y = 4) &= \sum_{n=4}^{\infty} P(Y = 4|X = n)P(X = n) = \sum_{n=4}^{\infty} \binom{n}{4} \left(\frac{2}{3} \right)^{n-4} \left(\frac{1}{3} \right)^4 \left(e^{-\lambda} \frac{\lambda^n}{n!} \right) \\ &= \frac{4}{13} \sum_{n=4}^{\infty} \binom{n}{4} \left(\frac{2}{3} \right)^{n-4} \left(\frac{1}{3} \right)^4 \left(\frac{\ln \left(\frac{13}{4} \right)^n}{n!} \right). \end{aligned}$$

3. Consider the probability space of choosing a point uniformly at random in the triangle with vertices at $(0, 0)$, $(1, 1)$ and $(2, 0)$.

(a) (5 points) Compute the marginal distribution of the y -coordinate $f_Y(y)$.

Solution: The triangle is bounded by the lines $y = 0$, $y = x$ and $y = 2 - x$. Also it has area one so $f(x, y) = 1$. Therefore

$$f_Y(y) = \int_{x=y}^{x=2-y} f(x, y) dx = 2 - 2y$$

(b) (5 points) Compute the expectation $E(XY)$.

Solution: Using the same x bounds as before we have:

$$E(XY) = \int_0^1 \int_y^{2-y} xy dx dy = \int_0^1 (2y - 2y^2) dy = 1/3$$

4. In this problem consider independent exponentials $X = \text{Exp}(\mu)$ and $Y = \text{Exp}(\lambda)$.

(a) (5 points) Compute the probability $\mathbb{P}(X < Y)$

Solution:

$$X \sim \text{Exp}(\mu), \quad Y \sim \text{Exp}(\lambda), \quad \text{independent}$$

We compute:

$$\mathbb{P}(X < Y) = \int_0^\infty \mathbb{P}(Y > x) f_X(x) dx$$

Since $Y \sim \text{Exp}(\lambda)$,

$$\mathbb{P}(Y > x) = e^{-\lambda x}, \quad f_X(x) = \mu e^{-\mu x}$$

Thus,

$$\begin{aligned} \mathbb{P}(X < Y) &= \int_0^\infty e^{-\lambda x} \mu e^{-\mu x} dx = \mu \int_0^\infty e^{-(\mu+\lambda)x} dx \\ &= \mu \cdot \frac{1}{\mu + \lambda} = \frac{\mu}{\mu + \lambda} \end{aligned}$$

(b) (3 points) Compute the PDF of $V = \min\{X, Y\}$

Solution: We have that

$$\{V > v\} = \{X > v\} \cap \{Y > v\}$$

. By independence and using the probabilities for exponential tails

$$P(V > v) = e^{-\mu x} e^{-\nu y}$$

Taking the complement probability $1 - e^{-\mu x} e^{-\nu y}$ we get the CDF. Then differentiating:

$$f_V(v) = (\mu + \nu)e^{-(\mu+\nu)v}$$

(c) (2 points) Find a function f such that $f(X)$ has the same distribution as Y . **Hint:** Instead of dealing with the CDF you can compare $\mathbb{P}(X > t)$ and $\mathbb{P}(Y > t)$.

Solution:

If we set

$$P(X > t) = e^{-\mu t} \quad P(Y > t) = e^{-\nu t}$$

$f(x) = \frac{\nu}{\mu}x$ works.

5. Consider an urn which contains three green balls and one red ball. You play a game where, each round, two balls are drawn without replacement. You win \$3 if one of the sampled balls is red, and otherwise you lose \$1.
- (a) (2 points) What is the probability p that you win money in a given round?

Solution: It's stated that you win \$3 when, having drawn two balls, the event {one of the sampled balls is red} occurs. (Note that this is not compatible with an interpretation that the reward is \$3 or -\$1 per ball.) We compute the probability of a win as follows:

$$p = P\{\text{one of the sampled balls is red}\} = \frac{\binom{1}{1}\binom{3}{1}}{\binom{4}{2}} = \frac{1 \times 3}{\frac{4 \times 3}{2 \times 1}} = \frac{1}{2}.$$

In other words, $p = P(X = 1)$ where $X \sim \text{Hypergeom}(4, 1, 2)$.

- (b) (2 points) Consider your total winnings W_n over n rounds. Using your answer to (a), find a function $g(s)$ such that $W_n = g(S_n)$, where S_n is a binomial random variable representing the number of rounds in which you won money out of n tries. *Hint: $g(s)$ can depend on n .*

Solution: We either win \$3 or lose \$1. Thus, if S_n is our number of successes out of n tries, our winnings after n rounds will be

$$W_n = 3S_n - (n - S_n) = 4S_n - n.$$

So the requested function is $g(s) = 4s - n$.

- (c) (5 points) What are the expectation $E(W_{100})$ and variance $\text{Var}(W_{100})$ of your total winnings after 100 rounds? Using these values and the normal approximation for the binomial distribution, estimate the probability you break even, i.e. that $W_{100} \geq 0$, after 100 rounds.

Solution: Given the relationship $W_n = 4S_n - n$, we can write the expectation and variance as

$$E(W_n) = 4E(S_n) - n, \quad \text{Var}(W_n) = 4^2 \text{Var}(S_n).$$

Because $S_n \sim \text{Bin}(n, \frac{1}{2})$, we can compute

$$E(S_{100}) = np = 100 \times \frac{1}{2} = 50, \quad \text{Var}(S_{100}) = np(1-p) = 100 \times \frac{1}{2} \times \frac{1}{2} = 25,$$

so that we can plug in to the above and obtain

$$E(W_{100}) = 4 \times 50 - 100 = 100, \quad \text{Var}(W_{100}) = 4^2 \times 25 = 400 = 20^2.$$

A normal approximation for $P(W_{100} \geq 0)$ can be obtained as

$$P(W_{100} \geq 0) = P(4S_{100} - 100 \geq 0) = P(S_{100} \geq 25) = P\left(\frac{S_{100} - 50}{5} \geq \frac{-25}{5}\right) \approx P(Z \geq -5).$$

It was also acceptable to write

$$P(W_{100} \geq 0) = P\left(\frac{W_{100} - 100}{20} \geq -\frac{100}{20}\right) \approx P(Z \geq -5),$$

since W_{100} is a linear function of the normally approximated S_{100} . Finally, $P(Z \geq -5) = \Phi(5) \approx 1$.

- (d) (3 points) Consider the effect of capping your winnings at an amount equal to $W_{\max} = E(W_{100})$. Let your reward after 100 rounds be $h(W_{100})$, where $h(w) = \min\{w, W_{\max}\}$. Estimate your expected reward $E[h(W_{100})]$ using the normal approximation.

You may use the fact that for a random variable $X \sim \text{Normal}(\mu, \sigma^2)$ with probability density $f_X(x)$,

$$\int_{-\infty}^{\mu} x f_X(x) dx = \frac{\mu}{2} - \frac{\sigma}{\sqrt{2\pi}}.$$

Solution: Applying the formula for expectation of a continuous random variable,

$$E[h(W_{100})] = \int_{-\infty}^{\infty} h(w) f_W(w) dw.$$

Approximate $f_W(w)$ as a normal pdf with $\mu = E(W_{100})$ and $\sigma^2 = \text{Var}(W_{100})$, since W_{100} is a linear function of the binomial S_{100} . Then $h(w) = \min\{w, \mu\}$, and we can split the above integral as

$$E[h(W_{100})] = \int_{-\infty}^{\mu} w f_W(w) dw + \int_{\mu}^{\infty} \mu f_W(w) dw = \left(\frac{\mu}{2} - \frac{\sigma}{\sqrt{2\pi}}\right) + \frac{\mu}{2} = \mu - \frac{\sigma}{\sqrt{2\pi}} = 100 - \frac{20}{\sqrt{2\pi}}$$

where the first integral is evaluated with the provided formula, and the second using $\int_{\mu}^{\infty} f_W(w) dw = \frac{1}{2}$.

6. A town of N people, all of whom enjoy card games, is divided into those who prefer sheephead (N_S), euchre (N_E), and cribbage (N_C) such that $N_S + N_E + N_C = N$. Denote the proportion of the town who enjoy each game as p_S , p_E , and p_C , where $p_S = N_S/N$, etc.

In a poll, k of the townspeople are asked to name their favorite card game. Assume $N \gg k$, so that you may approximate polling as sampling with replacement. Let (S_S, S_E, S_C) denote a random vector representing the number of poll responses for each of the three games.

- (a) (3 points) What is the joint distribution $P(S_S = k_S, S_E = k_E, S_C = k_C)$?

Solution: The distribution will be multinomial:

$$P(S_S = k_S, S_E = k_E, S_C = k_C) = \binom{k}{k_S, k_E, k_C} p_S^{k_S} p_E^{k_E} p_C^{k_C} = \frac{k!}{k_S! k_E! k_C!} p_S^{k_S} p_E^{k_E} p_C^{k_C}.$$

- (b) (5 points) For $k = 900$ and $S_S = 300$, consider the apparent sheephead preference $\hat{p}_S = S_S/k$. Find the 95% confidence interval for p_S , i.e. find ε such that $P(|\hat{p}_S - p_S| < \varepsilon) \geq 0.95$.

Solution: The marginal distribution of S_S is binomial with k trials and success probability p_S , so our apparent preference $\hat{p}_S = 300/900 = \frac{1}{3}$. To find a confidence interval, we write

$$P(|\hat{p}_S - p_S| < \varepsilon) \geq 2\Phi(2\varepsilon\sqrt{k}) - 1 \geq 0.95.$$

Noting $\sqrt{k} = \sqrt{900} = 30$ and re-arranging,

$$\Phi(60\varepsilon) \geq 0.975 \implies 60\varepsilon \geq 2 \implies \varepsilon \geq \frac{1}{30}.$$

Note that we used the closest entry on the table of $\Phi(z)$ values, $\Phi(2.0) = 0.977$. Finally, this suggests our 95% confidence interval is

$$\left(\frac{1}{3} - \frac{1}{30}, \frac{1}{3} + \frac{1}{30}\right) = \left(\frac{3}{10}, \frac{11}{30}\right) \approx (0.3, 0.3667).$$

List of named distributions

$$X \sim \text{Bernoulli}(p), \quad \text{PMF: } p_X(0) = 1 - p, \quad p_X(1) = p$$

$$X \sim \text{Binomial}(n, p), \quad \text{PMF: } p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad 0 \leq k \leq n$$

$$X \sim \text{Geometric}(p), \quad \text{PMF: } p_X(k) = p(1 - p)^{k-1}, \quad k = 1, 2, \dots$$

$$X \sim \text{Poisson}(\lambda), \quad \text{PMF: } p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \dots$$

$$X \sim \text{Hypergeometric}(N, N_A, n), \quad \text{PMF: } p_X(k) = \frac{\binom{N_A}{k} \binom{N - N_A}{n - k}}{\binom{N}{n}}, \quad 0 \leq k \leq n$$

$$X \sim \text{Uniform}(a, b), \quad \text{PDF: } f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$$

$$X \sim \text{Normal}(\mu, \sigma^2), \quad \text{PDF: } f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$$

$$X \sim \text{Exponential}(\lambda), \quad \text{PDF: } f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$$

$$X \sim \text{Gamma}(r, \lambda), \quad \text{PDF: } f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$$

$$(X_1, \dots, X_k) \sim \text{Multinomial}(n, p_1, \dots, p_k),$$

$$\text{joint PMF: } p(a_1, \dots, a_k) = \binom{n}{a_1, \dots, a_k} p_1^{a_1} \cdots p_k^{a_k} \text{ for } a_1, \dots, a_k \geq 0 \text{ with } a_1 + \dots + a_k = n.$$

The following table shows approximate values of the cumulative distribution function of the standard normal distribution.

x	$\Phi(x)$
-2.0	0.022
-1.9	0.028
-1.8	0.035
-1.7	0.044
-1.6	0.054
-1.5	0.066
-1.4	0.080
-1.3	0.096
-1.2	0.115
-1.1	0.135
-1.0	0.158
-0.9	0.184
-0.8	0.211
-0.7	0.242
-0.6	0.274
-0.5	0.308
-0.4	0.344
-0.3	0.382
-0.2	0.420
-0.1	0.460
0.0	0.500
0.1	0.539
0.2	0.579
0.3	0.617
0.4	0.655
0.5	0.691
0.6	0.725
0.7	0.758
0.8	0.788
0.9	0.815
1.0	0.841
1.1	0.864
1.2	0.884
1.3	0.903
1.4	0.919
1.5	0.933
1.6	0.945
1.7	0.955
1.8	0.964
1.9	0.971
2.0	0.977